CURRICULUM VITAE

Belal Zaky

MSci Pharmacology, King's College London · London | Dubai

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EDUCATION

Expected 2029 — **King's College London — MSci Pharmacology (integrated master's, with placement year)**
London, United Kingdom

2024 — **Emirates International School – Jumeirah — IB Diploma**
Dubai, United Arab Emirates

SELECTED PROJECTS

2026 — **Causal safety — does an NSAID cause kidney injury?**

Python · propensity-score matching · self-controlled case series · bootstrap CI

- Emulated a new-user, active-comparator study and validated the estimators against a sealed true effect: the crude comparison understated the harm by 46% (OR 1.19), while matching (2.05) and a self-controlled series (2.14) both recovered the truth (2.20).
- A simulation study by design — the only setting where you can score a causal method against a known answer. Same protocol queued for real clinical data (MIMIC-IV, credentialing in progress).

2026 — **PV Intake Assistant — production pharmacovigilance pipeline**

Python · scikit-learn · scispaCy · SNOMED CT · pytest · GitHub Actions · Docker

- Three-stage pipeline: triage a medical abstract for a possible adverse drug event (F1 0.38 → 0.79), extract the drug and effect, then code the effect to SNOMED CT against a 461,000-term vocabulary.
- Built as a system, not a notebook — 32 unit tests, CI on every push, a Dockerfile and a CLI. Chose precision over coverage in coding: a wrong code corrupts a safety signal, a blank one does not.

2026 — **Pharmacovigilance signal detection — FAERS (20M+ FDA reports)**

Python · pandas · SQL · openFDA API · PRR / ROR / chi-squared / Evans criteria

- Built a disproportionality signal-detection pipeline over 20M+ adverse-event reports; validated on the known warfarin–haemorrhage signal (PRR ≈ 5), with negative controls to screen out confounders.
- Across companion projects (a live dashboard and a SQL reproduction), showed how notoriety and sampling distort conclusions — a heartburn drug's 2020–21 report spike was litigation, not clinical harm.

2026 — **Clinical NLP & EHR modelling — adverse events, risk and survival**

R (survival) · Kaplan–Meier · Cox PH · Python · scikit-learn · PubMedBERT

- Benchmarked four models for detecting adverse drug events in free text: the domain-tuned transformer won, a keyword model beat a general-purpose one, and a zero-shot LLM came last — domain fit beats model size.
- Modelled time-to-hypertension in 1,100+ patients: diabetics progressed ~2.4× faster, and a seemingly null age effect proved time-varying.

2026 — **Mission Control — my own operations app**

JavaScript · Electron · Node · Markdown · AI agents

- A desktop app that runs my projects, study and deadlines from plain-text notes, with an AI assistant whose drafts I approve; 600+ automated tests, each written after a real failure.

EXPERIENCE

Jul 2024	**Aster Pharmacy — Sales Intern** Supported customer service and sales operations in a busy retail pharmacy environment, Dubai.
Jun 2022 – Apr 2023	**Modon Designs — Project Coordinator & Business Development Associate** Coordinated client–team communication and documentation, built digital filing systems, and supported business development and social media.
Jan 2022 – Jun 2023	**M2 Supplements — Operations & Marketing Assistant** Ran order fulfilment, customer communication and social-media campaigns, supporting product launches with the founders.

LEADERSHIP & COMMUNICATION

2025 – Present	**Student Representative — Pharmacology, King's College London** Rep in Year 1 (BSc) and again in Year 2 (MSci). Represent coursemates on the Staff–Student Liaison Committee; consolidated scattered module feedback into one case that reached the department and changed the module.
2026	**King's100 — Leadership & Employability Programme** Selected for King's flagship leadership programme — employer-led challenges.
2023 – Present	**Toastmasters International — Member & Sergeant at Arms** Regular prepared and impromptu speaking; earned "Speech of the Meeting".
2023	**Model United Nations — Delegate** Awarded Delegate of the Conference — research, debate and negotiation.

SKILLS

Data & technical	Python (pandas, scikit-learn, matplotlib), R (survival analysis), SQL, machine learning, NLP; Git/GitHub, pytest, GitHub Actions CI, Docker, Astro
Methods	Causal inference (propensity scores, self-controlled case series), survival analysis, disproportionality signal detection, statistical reasoning
Domain	Pharmacovigilance & drug-safety signal detection, real-world evidence, EHR data, SNOMED CT coding
Communication	Scientific & report writing, data storytelling, public speaking. English (fluent), Arabic (native)